CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

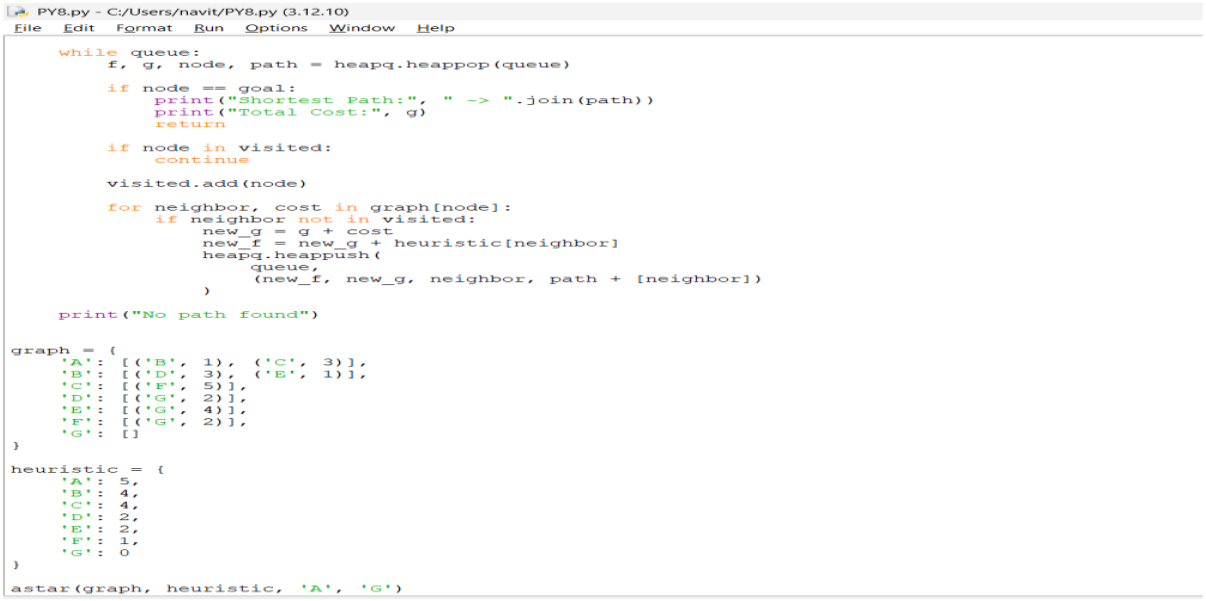
Program 10: WRITE THE PYTHON POGRAM TO IMPLEMENT A* ALGORITHM

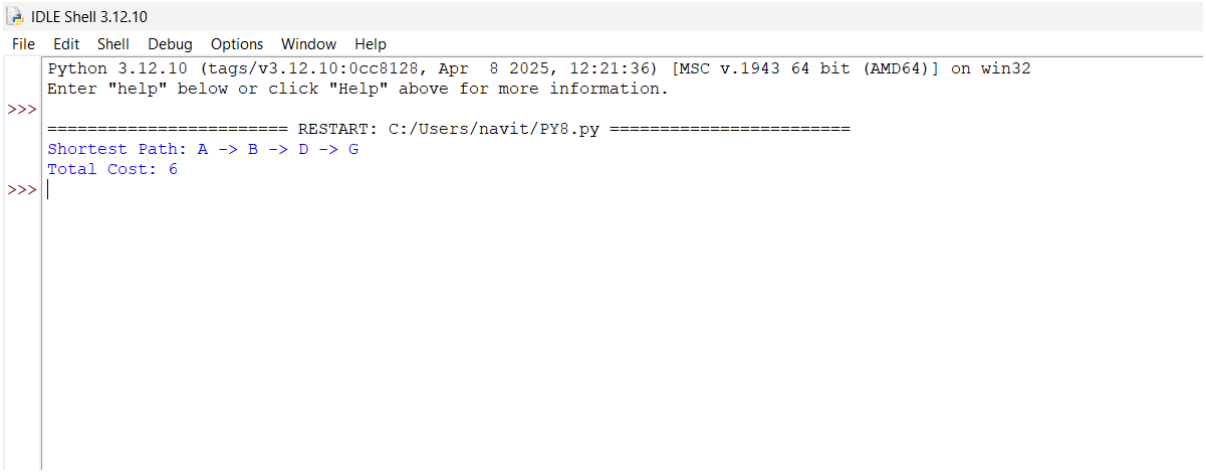
Aim:

To write a Python program to implement the A* (A-Star) algorithm for finding the shortest path between a start node and a goal node using path cost and heuristic cost.

Algorithm:

1. Start the program.
2. Define the graph with edge costs and heuristic values.
3. Initialize the starting node with a path cost of zero.
4. Calculate the evaluation function: **f(n) = g(n) + h(n)**.
5. Select the node with the lowest evaluation cost.
6. Expand the selected node and calculate the costs of its neighboring nodes.
7. Update the path costs and parent nodes when a better path is found.
8. Repeat until the goal node is reached or no path remains.
9. Display the shortest path and its total cost.
10. Stop the program.





Result:

The Python program was executed successfully, and the A* algorithm found the shortest path from the start node to the goal node.